

Can MLLMs Reason in Multimodality? EMMA: An Enhanced MultiModal ReAsoning Benchmark

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Background: Multimodal AI is Transforming the World

- Humans solve STEM tasks by fusing diagrams, text, and symbols.
- MLLMs promise assistants for education, science, and visualization.
- True value requires integrated visual-text reasoning, not perception only.

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Outline

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The Problem: Can MLLMs Perform Organic Multimodal Reasoning?

- Current MLLMs excel at text-only reasoning; visuals remain shallow.
- Benchmarks often allow language-only or single-pass visual shortcuts.
- We need tasks that require intertwined visual-text, multi-step reasoning.

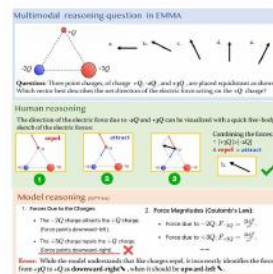


Figure: EMMA示例：人类作图合成电场力，GPT-4o误判斥力方向，凸显多模态推理局限。

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Why Existing Benchmarks Fall Short

- Redundant text–image info enables caption-only solutions.
- Tasks often test perception, not deep visual manipulation.
- Little emphasis on multi-step, cross-modal inference.

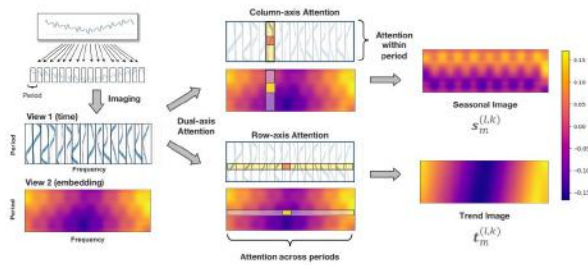


Figure: 无需图像即可作答，来自数学数据集Math-Vista与Math-Vision；Q/C/I分别表示问题/图注/图像

Core Idea & Contributions (Overview)

- EMMA: a benchmark targeting organic multimodal reasoning in STEM and coding.
- Strict filtering to remove text-only or single-pass visual solutions.
- Fine-grained skill labels for diagnosis and progress tracking.

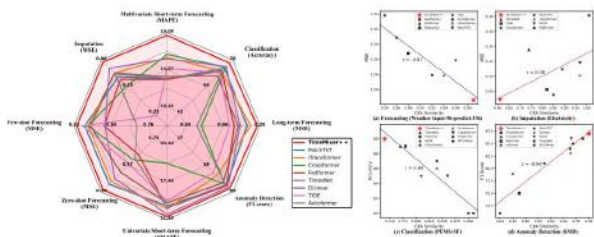


Figure: EMMA概览：涵盖数理化与编程；题目按技能分类，支持细粒度分析。

EMMA at a Glance: Composition and Scope

- 2,788 questions across math, physics, chemistry, coding.
- Multiple formats: MC and free-form; images in 93% of prompts.
- Balanced EMMA-mini (100 per subject) for fast, diagnostic runs.

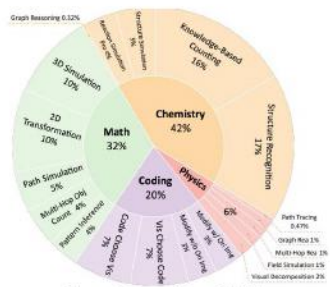


Figure: Composition of EMMA.

Dataset Statistics (Mandatory)

- Large, image-heavy, and diverse across subjects.
- 64% newly created to target cross-modal reasoning.
- 10% multi-image items to support multi-hop visual inference.

Table: Table 1: Key statistics of EMMA.

Statistic	Number
Total questions	2,788
- Multiple-choice questions	2,002 (72%)
- Free-form questions	786 (28%)
- Questions with answers	2,788 (100%)
- Questions newly added	1,796 (64%)
- Image in the question	2,500 (90%)

- Step 1: Caption-augmented filtering with strong LLMs, multi-trial.
- Discard if any model $\geq 5/10$ correct (text-only or shallow visual).
- Step 2: Author new items aligned to skill taxonomies; expert-verified.

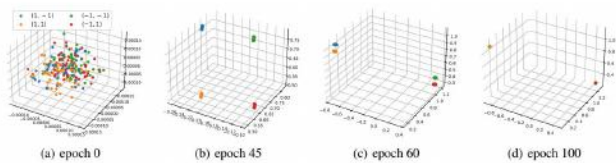


Figure: Data curation process for EMMA (Sec. 3.2).

- Taxonomies per subject: 2D/3D simulation, path tracing, graphs, coding skills.
- Labels support targeted evaluation and failure diagnosis.
- Taxonomy aligns with external datasets for cross-benchmark analyses.

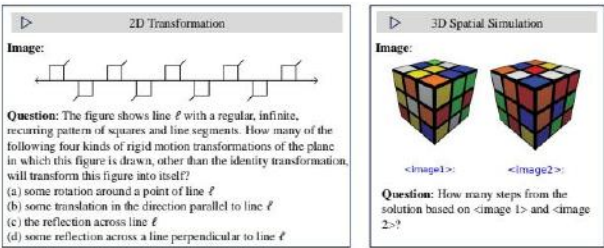


Figure: MMMU两例(Yue et al., 2024a): 分别属2D变换与3D空间模拟; 分类适用于其他数据集

- Rotation, translation, reflection—mental manipulation required.
- Tests spatial imagination beyond textual heuristics.
- CoT can harm when over-verbalizing instead of simulating.

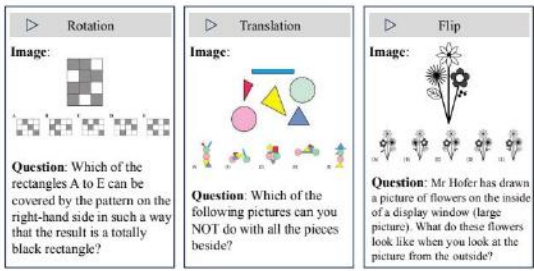


Figure: 数学中二维变换的三类问题: 平移、旋转、反射

- Trajectory following and perspective shifts.
- Requires iterative, stepwise visual reasoning.
- Hard for models without visual scratchpads.

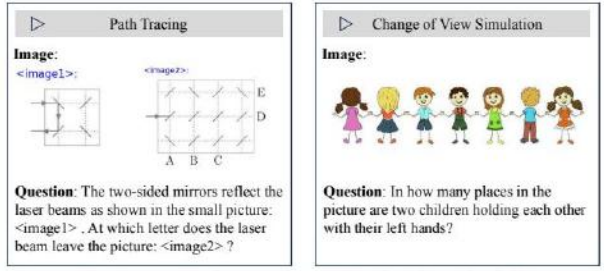


Figure: 数学中路径追踪/视角变换模拟的两个典型例子

- Filter by attributes, then compute over subsets.
- Pattern abstraction across panels (e.g., RAVEN).
- Labels isolate skill-specific weaknesses.



Figure: 数学多跳视觉目标计数的四个典型问题

- Structure recognition and bond counting from diagrams.
- Arrow-pushing, mechanism simulation (incl. pro-level).
- Predominantly new items with expert verification.

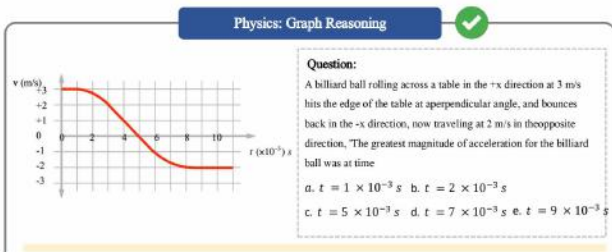


Figure: A sample correct case in physics.

- Four tasks: Vis→Code, Code→Vis, Modify w/ and w/o image.
- All multiple-choice—no LLM judge dependency.
- Skill tags: axis/scale, legend, 3D, advanced charts.

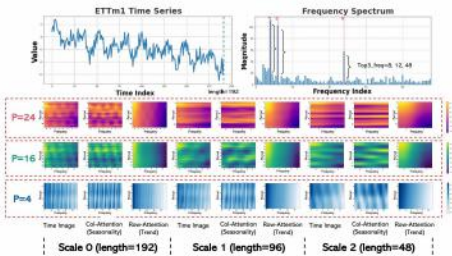


Figure: 相较重建式基准，EMMA以四类选择题评估可视化编码，规避主观判分并细粒度剖析多模态编码局限

- Full EMMA + balanced EMMA-mini (100 per subject).
- Human-expert baseline; 10 SOTA MLLMs (open/closed).
- Direct vs CoT; test-time scaling (N up to 16).

Main Results: Large Human–Model Gap

- Humans: 77.75% on EMMA-mini.
- Best model: 48.00% (−29.75% vs humans).
- Some models near random on hard subsets.

Table: Table 2: Evaluation results of state-of-the-art MLLMs, which are outperformed by human experts with wide margins. The highest and second-highest scores are highlighted in green and blue , respectively. ↑ indicates CoT improvements, while ↓ denotes reductions.

Model	CoT	EMMA					EMMA-mini				
		Math (892)	Phys. (156)	Chem. (1,176)	Coding (564)	Overall (2,788)	Math (100)	Phys. (100)	Chem. (100)	Coding (100)	Overall (400)
Random choice	-	14.01	25.64	16.50	25.71	18.08	13.00	23.00	27.00	28.00	22.75
Human Expert	-	-	-	-	-	-	75.00	64.50	86.00	85.50	77.75
Claude 3.5 Sonnet	×	25.34	33.97	40.90	38.65	35.08	23.00	34.00	44.00	35.00	34.00
Gemini 2.0 Flash	×	23.88	38.46	36.31	42.02	33.61	20.00	40.00	36.00	41.00	34.25
GPT-4o	×	27.24	38.46	31.89	40.07	32.42	30.00	38.00	33.00	40.00	35.25
Qwen2-VL-72B-Instruct	×	33.07	42.31	32.06	34.57	33.46	38.00	40.00	34.00	37.00	37.25
LLaVA-Onevision-72B	×	27.69	35.90	25.26	28.72	27.33	25.00	32.00	24.00	28.00	27.25
InternVL2-Llama3-76B	×	25.11	22.44	24.06	27.84	25.07	31.00	22.00	21.00	28.00	25.50
InternVL2.5-78B	×	31.39	38.46	35.20	31.91	33.50	30.00	40.00	38.00	33.00	35.25
Claude 3.5 Sonnet	✓	29.37	41.03	41.07	40.60	37.23 (↑ 2.15)	30.00	38.00	41.00	39.00	37.00 (↑ 3.00)
Gemini 2.0 Flash	✓	25.90	38.46	24.66	40.96	29.12 (↓ 4.48)	24.00	41.00	36.00	44.00	36.25 (↑ 2.00)
GPT-4o	✓	25.56	43.59	33.67	39.01	32.71 (↑ 0.29)	27.00	44.00	35.00	38.00	36.00 (↑ 0.75)
Qwen2-VL-72B-Instruct	✓	27.69	34.62	24.57	29.43	27.12 (↓ 6.35)	35.00	34.00	32.00	23.00	31.00 (↓ 6.25)
LLaVA-Onevision-72B	✓	22.42	15.38	22.70	30.67	23.82 (↓ 3.52)	23.00	26.00	23.00	29.00	25.25 (↓ 2.00)
InternVL2-Llama3-76B	✓	22.20	32.05	19.73	30.32	23.35 (↓ 1.72)	27.00	33.00	21.00	32.00	28.25 (↑ 2.75)
InternVL2.5-78B	✓	25.56	39.74	27.47	25.18	27.08 (↓ 6.42)	31.00	36.00	24.00	19.00	27.50 (↓ 7.75)
Gemini 2.0 Flash Thinking 1210	✓	31.61	56.41	27.02	42.44	38.06	35.00	57.00	41.00	41.00	42.50

Test-Time Compute Scaling: Gains but Still Far from Humans

- Majority voting, Best-of-N, tournaments up to N=16.
- Modest gains: best still ≈27% below humans.
- Independent rewarders outperform self-reward.

Table: Table 3: Results of different test-time scaling strategies on EMMA-mini, including (1) majority voting, (2) best-of-N, (3) tournamentstyle best-of-N. Pass@N accuracies are included as upper bounds of the scaling performance.

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	-		-	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	-
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.50	-
	Tournament	Gemini Flash Thinking		40.75	39.25	41.25	35.25
	Pass@N	-		45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	-		-	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	-
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	-
	Tournament	Gemini Flash Thinking		36.75	37.25	40.75	38.75
	Pass@N	-		45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	-		-	48.00	49.00	50.75
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	-		53.75	64.50	71.50	81.50

CoT vs Direct Prompting: Task-Dependent Effects

- Closed-source often benefit from CoT; open-source may degrade.
- Visual-simulation-heavy tasks hurt with textual CoT.
- Language-friendly tasks (e.g., multi-hop counting) can improve.

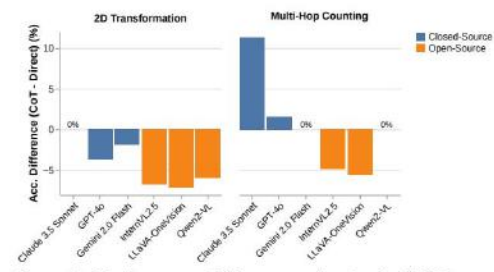


Figure: 有/无CoT在两项数学任务（二维变换、多跳目标计数）上的性能差异

Diagnostic Error Analysis: Visual Reasoning Bottleneck

- Visual reasoning errors dominate (~53%).
- Perceptual errors are secondary (~30%).
- Textual/knowledge errors are much smaller.

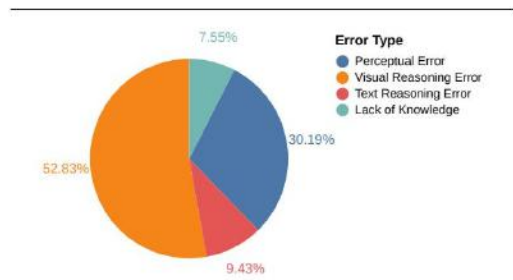


Figure: o1在EMMA-mini数学与编程部分的错误类型分布：视觉推理为主要来源

- Text-only chains can impose wrong spatial narratives.
- CoT sometimes flips correct answers to wrong ones.
- Visual simulation is needed, not just longer text.

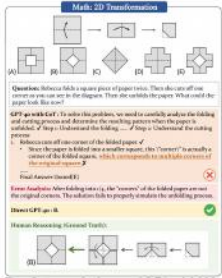


Figure: 文本CoT不利于求解: GPT-4o无CoT正确, 有CoT错误, 反映表层空间联想非真正视觉推理。

- Models recall rules but fail 3D hand orientation.
- Spatial imagination gaps cause systematic errors.
- Indicates need for native visual-manipulation modules.

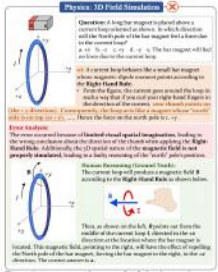


Figure: o1视觉推理失误: 因空间模拟受限误用右手定则

- STEM education: robust diagram-based problem solving.
- Scientific practice: graphs, mechanisms, spatial models.
- Visualization workflows: Vis↔Code mapping and debugging.

- Manual curation biases; physics coverage smaller.
- Chemistry focus could broaden; automate curation.
- Responsible deployment: avoid deceptive visual misuse.

- Rigorous, skill-labeled benchmark for multimodal reasoning.
- Visual reasoning is the bottleneck; text CoT is insufficient.
- Compute scaling helps modestly; far from human level.

- Architectures for visual simulation/manipulation.
- Visual scratchpads/tools and visual CoT strategies.
- Better multimodal reward models for selection.

- Thank you for your attention!
- Questions and feedback are welcome.

- Thanks to collaborators, annotators, and domain experts.
- Appreciation for responsible AI discussions in the community.